

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of Bachelor of Optometry Examination April 2018

BOP204 Physical Optics

Date: 18/04/2018 Day: Wednesday Time: 10:00 am To 10.30 am Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I (A) Choose the correct answer for the following questions. 10

1. Polarization phenomena can be observe by.....of light
 - (a) Wave Nature
 - (b) Particle Nature
 - (c) a & b both
 - (d) None of the above
2. If direction of particle oscillation (of electromagnetic field) in light is parallel to the direction of propagation of light is called,
 - (a) Transverse Nature
 - (b) Longitudinal Nature
 - (c) Quantum Nature
 - (d) None of the above
3. Electromagnetic vibration which are to the alignment of the molecules is absorb in polarizer.
 - (a) Perpendicular
 - (b) Parallel
 - (c) 45°
 - (d) None of the above
4. At which angle between polarized and analyzer the passing wave has a highest intensity and magnitude.
 - (a) 60
 - (b) 50
 - (c) 30
 - (d) 40

5. Ray obey the normal law of refraction and travels with the same velocity in every direction within the crystal and is called,
- (a) Incident ray (b) Extraordinary ray
(c) Ordinary ray (d) Reflected ray
6. Phenomenon of merging waves having same frequency is called.....
- (a) Interference (b) Polarization
(c) Coherence (d) Refraction
7. For first order dark fringe the path difference between two wave is.....
- (a) 2λ (b) λ
(c) $\frac{3\lambda}{2}$ (d) $\lambda/2$
8. In fluorescence process the wavelength of emitted light is then absorb light.
- (a) Shorter (b) Longer
(c) Equal (d) None of the above
9. Radiant energy emitted, reflected, transmitted or received per unit time is called.....
- (a) Radiant intensity (b) Radiance
(c) Irradiance (d) Radiant flux
10. Illuminance measured at 2-meter distance from the source will be..... of the measure at 1 meter.
- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$
(c) $\frac{1}{16}$ (d) $\frac{1}{25}$

Q - I (B)	Match the following column "A" with Column "B".		5
Column A	Answer	Column B	
1. Double slit experiment		a. Resolution	
2. Rayleigh's Criterion		b. Lumen.sec	
3. Photoluminescence		c. Interference	
4. Luminous energy		d. Troland	
5. Retinal illuminance		e. Fluorescence	

Q - I (C)	Select whether following statement is true or false.	5
		True/False
1.	Wave nature of the light can't be understood by Interference phenomena.	
2.	$d\theta$ refers to resolving power of telescope..	
3.	X-rays are used to view inside the body.	
4.	Scattering of yellow light more than red light.	
5.	SI unit of solid angle is Troland.	

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Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION –I

Q - II Answer the following questions as directed. (Any ten)

20

1. Define Wavelength and Amplitude in term of sine wave.
2. Define Transverse and Longitudinal waves.
3. Define Diffraction grating with its work.
4. Differentiate between Radio waves and Gamma rays. (two points)
5. Write Rayleigh criteria for resolution.
6. Define and derive the equation of Brewster's angle.
7. Define Ordinary and Extraordinary ray in Birefringence
8. Define Fluorescence and Phosphorescence process.
9. Define luminous flux and luminous intensity with proper units.
10. Define retinal illuminance with equation.
11. Derive the relationship between intensity and amplitude.

SECTION – II

Q-III Answer the following questions as directed. (Any four)

20

1. Explain Coherence and Interference.
2. Write short note on Natural and Artificial light source.
3. Write short note on Radiometric units.
4. Derive the equation for resolving power of telescope.
5. Explain Young's double slit experiment.

Q-IV Answer the following questions as directed. (Any one)

10

1. Explain solid angle, Inverse square law and Lambert law.
2. What is Polarization? Explain polarization by Transmission and Reflection.